

Ali Nikkhah

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SUMMARY

Computer Vision Engineer with production experience across edge deployment, multimodal retrieval, and audio-visual intelligence systems. At Advanced Analytics Australia, delivered end-to-end surveillance analytics on AXIS edge hardware using TensorRT, Triton Inference Server, and Apache Kafka for high-throughput stream management. At Digikala, led multimodal product search using Vision Transformers (ViT) and BLIP across millions of product assets. Research background includes SOTA results on automated ultrasound report generation (UBC, BLEU/ROUGE-L) and texture-free video motion classification (L3S, UCF-101/HMDB-51).

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript
Deep Learning & AI: PyTorch, TensorFlow, JAX, Hugging Face, LangGraph, LangChain, LlamaIndex, FAISS, HyDE, KAG
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, TensorRT, TensorFlow Lite
Data & Streaming: Spark, Trino, Airflow, Kafka, PeerDB, Elasticsearch, Parquet
MLOps & Infra: Docker, Kubernetes, Terraform, Ray, Triton, vLLM, MLflow, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Advanced Analytics Australia

Remote Contract

Computer Vision Engineer

Sep 2025 – Apr 2026

- Delivered production video analytics on AXIS edge hardware for industrial safety compliance across multiple sites: handwash verification, PPE detection (boots, helmets), face tracking, and restricted-zone entry alerting — all running on-device without cloud round-trips.
- Compiled and quantized vision models into **TensorRT** and **TensorFlow Lite** runtimes; served them via **Triton Inference Server** with dynamic batching to eliminate Python-level inference overhead on resource-constrained edge nodes; managed concurrent high-frame-rate camera streams through a distributed **Apache Kafka** metadata broker.
- Implemented dynamic sensor calibration routines compensating for exposure variance and lighting shifts across indoor/outdoor placements, maintaining stable detection thresholds at all deployment sites.

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led multimodal product retrieval using **Vision Transformers (ViT)** and **BLIP** encoders to align images, diagrams, and text in a shared vector space, enabling cross-modal text-to-image semantic search across millions of product listings.
- Built high-throughput image preprocessing workers fully decoupling raw upload ingestion from model inference; profiled and optimized matrix operations to eliminate OOM failures under heavy concurrent load; supervised multi-tenant Kubernetes deployment isolating vision serving nodes from language generation clusters.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Scientist & AI Engineer

Sep 2025 – Mar 2026

- Built a real-time telephony voice intelligence pipeline: live customer call audio captured and transcribed via **Whisper** and **wav2vec2.0**, then routed into a LangGraph agent for automated support ticket creation and triage.
- Compiled acoustic feature extraction to **TorchScript**; built Asyncio workers processing parallel voice payloads without blocking API threads.

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Orchestrated model training and data pipelines on Kubernetes (Ray, MLflow); built low-latency FastAPI inference APIs with TensorFlow Serving; automated CI/CD with Docker, ArgoCD, and GitHub Actions, cutting release cycles to under 10 minutes.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024

RESEARCH HIGHLIGHTS

Univ. of British Columbia (*Remote RA, Feb 2024 – Nov 2025*) — Vision-language architectures (ViT + BLIP + T5) with self-supervised contrastive learning for automated DICOM-compliant ultrasound report generation; achieved SOTA BLEU & ROUGE-L. Preprint in preparation.

L3S Research Center (*Remote RA, Apr 2023 – Feb 2024*) — 3D CNNs and transformer encoders over dense optical flow for texture-free motion classification; SOTA on UCF-101 and HMDB-51.